Differentiable Hybrid Surrogates for Learning, Optimization, and Control of Power-Grid Dynamics

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I. Abstract

High-fidelity simulation of transmission-grid dynamics is essential for evaluating fault responses, but its computational cost limits its use in repeated simulation, optimization, and control. Reduced-order models are substantially faster but may omit dynamics that become important under changing operating conditions. This proposal develops a differentiable hybrid surrogate that combines known power-system dynamics with learned corrections for unresolved effects, following the physics-constrained and differentiable modeling paradigm of [1], [2].

The work builds directly on my doctoral research, where I developed a stochastic surrogate based on linearized swing dynamics and calibrated its parameters against high-fidelity EMT simulations. For the IEEE 9-bus benchmark, the existing model reproduces the phase-angle configuration at fault clearing with a mean squared relative error below 4.5% for fault durations up to 1.5 s and a maximum observed error of 9% over the tested contingencies. This provides a quantitative starting point for learning the remaining model discrepancy rather than replacing the physical model.

The central hypothesis is that a topology-aware differentiable surrogate, in which known swing dynamics are augmented by a learnable graph-based correction, can improve predictive accuracy while preserving physical structure and enabling gradients through the simulated dynamics. The resulting model will support three coupled tasks: physics-informed system identification, differentiable dynamic-security optimization, and model-based predictive control. This organization follows the learning-to-model, learning-to-optimize, and learning-to-control framework developed in NeuroMANCER [3], [4], [5]. The outcome will be an open-source differentiable modeling framework for transferring information from expensive high-fidelity simulations into computationally efficient models for power-system decision and control problems.

II. Introduction

Modern transmission grids combine increasingly heterogeneous generation, reduced inertia, variable injections, and tightly coupled network dynamics. These features make the response to a transmission fault dependent not only on the pre-fault operating point but also on the transient evolution of the system. High-fidelity EMT models can resolve these dynamics, but their computational cost makes exhaustive simulation across contingencies, operating conditions, and control decisions impractical.

This creates a model-reduction problem: how can information contained in high-fidelity simulations be transferred to a computational model without discarding the physical structure needed for reliable extrapolation? My doctoral research addresses this problem through a stochastic surrogate based on linearized swing dynamics, explicit fault-induced changes in network coupling, and stochastic forcing for unresolved effects [6], [7]. The model has already been calibrated against EMT trajectories and provides a substantially cheaper representation of the transient phase-angle response.

The proposed research extends this surrogate rather than replacing it. The physical swing dynamics and network topology will remain explicit, while a learnable correction will represent systematic discrepancies between the reduced model and EMT dynamics. Because the transmission network is naturally a graph, the initial model class will use a topology-aware graph neural network embedded within a neural differential-equation formulation. A simpler neural residual will serve as a baseline. This structure is consistent with the universal differential equation framework, in which mechanistic models are augmented with machine-learnable components [2].

The resulting differentiable model will provide a common computational representation for three tasks: learning the unresolved dynamics from EMT data, optimizing operating and contingency decisions through the model, and computing feedback control policies using the same differentiable dynamics. These tasks share a common driving force: gradients through the hybrid dynamical model. This provides the basis for the proposed two-year research program.

III. Background and Technical Foundations

The electromechanical response of a transmission network following a disturbance is described, to leading order, by the swing equations, which couple phase angles, frequencies, power injections, and network topology [8]. After linearization around an operating point, network coupling is represented by a Laplacian-based dynamical operator. A transmission fault modifies this operator during the fault interval, providing an explicit representation of the disturbance.

My current surrogate has the form

$$dx(t) = [A_e(\alpha_e)x(t) + P]\,dt + G_e(\sigma_e)x(t)\,dW(t),$$

where α_e is a scalar parameter controlling the effective fault strength on line e and σ_e is a scalar stochastic-forcing amplitude. The model separates known fault-induced dynamics from unresolved variability and is calibrated against EMT trajectories.

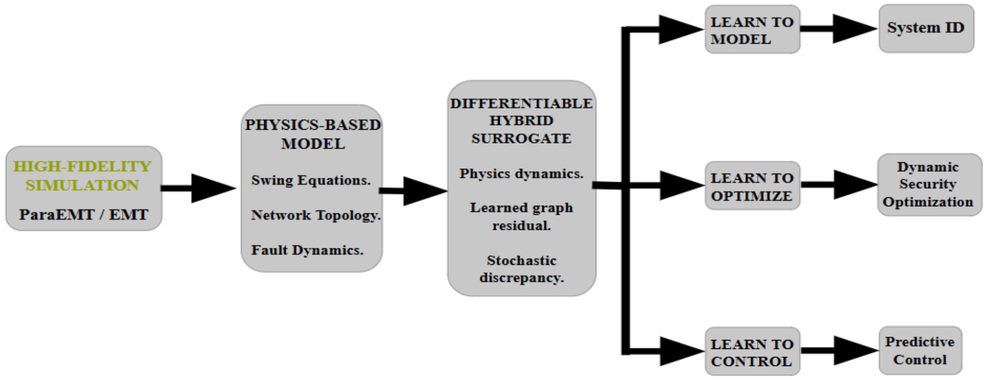

Fig. 1: Conceptual workflow of the proposed differentiable hybrid surrogate. High-fidelity EMT simulations provide data for a physics-based representation that is augmented with learned unresolved dynamics. The resulting differentiable surrogate serves as a common computational model for system identification, dynamic-security optimization, and predictive control.

This existing surrogate model θ^{surr} provides a quantitative foundation for the proposed extension. Let $\theta^{\text{EMT}}(\tau)$ and $\theta^{\text{surr}}(\tau)$ denote the phase-angle states at fault clearing obtained from ParaEMT [9] and surrogate respectively, here $\tau \in \mathbb{R}^+$ represents the tripped line's re-energization instant. The validation metric

$$\text{RSE}(\tau) = \frac{\mathbb{E}\big[\|\theta^{\text{EMT}}(\tau) - \theta^{\text{surr}}(\tau)\|^2\big]}{\mathbb{E}[\|\theta^{\text{EMT}}(\tau)\|^2]}$$

remains between the empirical bounds 0.02τ and 0.0425τ across the nine tested line faults, with a maximum observed value of 9% and values below 4.5% for $\tau \leq 1.5$ s. These results indicate that the existing physical surrogate captures the dominant phase-angle configuration relevant to the post-fault dynamics and provide a baseline against which learned corrections can be evaluated.

The proposed model will augment this structure as

$$\dot{x} = f_{\text{phys}}(x, u; \boldsymbol{\vartheta}) + f_{\text{GNN}}(x, \mathcal{G}, u; \boldsymbol{\phi}) + G(x; \boldsymbol{\psi})\, \xi.$$

Here $\boldsymbol{\vartheta} \in \mathbb{R}^p$ contains physical parameters, $\boldsymbol{\phi} \in \mathbb{R}^q$ contains the parameters of the learned graph-based correction, and $\boldsymbol{\psi} \in \mathbb{R}^r$ parameterizes the stochastic component. The first term f_{phys} is fixed by the swing-equation and network structure, the second f_{GNN} represents systematic model discrepancy, and the third G represents unresolved stochastic effects.

This is a universal-differential-equation formulation [2]. Its key advantage is that the learned component does not need to rediscover the known network physics. Instead, high-fidelity data are used to identify the discrepancy left by the reduced model. Automatic differentiation and sensitivity analysis then provide gradients with respect to model parameters, operating variables, and control policies. Adjoint-based sensitivities will be investigated for long-horizon optimization, building on recent differentiable simulation approaches for coupled energy systems [10].

IV. Proposed Research and Methodological Plan

The research will develop one differentiable hybrid surrogate and progressively use it for modeling, optimization, and control. The three objectives share the same learned dynamical representation but address different computational tasks. **Figure 1** summarizes this overall research architecture, while **Table I** summarizes its two-year implementation.

Objective 1: Learning to Model. The first objective is to convert the existing stochastic surrogate into a differentiable hybrid model. The swing equations, network topology, and fault-induced changes in network coupling will remain explicit. A topology-aware graph neural network will provide the initial representation of systematic model discrepancy, embedded within a neural differential-equation formulation [2]. A neural state-space residual will be used as a lower complexity baseline. EMT trajectories from ParaEMT will provide the reference data. Because these simulations are expensive and operate at a higher fidelity than the reduced model, data generation will be treated as an offline stage; trajectory alignment, fault-event synchronization, and separation of calibration and validation contingencies will be explicitly addressed. Model quality will be assessed using phase-angle RSE, trajectory error, computational cost, and generalization to unseen fault locations and operating points.

Objective 2: Learning to Optimize. The second objective is to use the differentiable surrogate to replace repeated high-fidelity simulations inside dynamic-security optimization. In contrast to control, the optimization problem will determine an operating or contingency decision before the resulting trajectory is realized. A representative problem is

$$\min_{\mathbf{u}} \; J(x_{0:T}, \mathbf{u}) \quad \text{s.t.} \quad x_{k+1} = S_\phi(x_k, u_k), \qquad g(x_{0:T}, \mathbf{u}) \leq 0.$$

where S_ϕ denotes the differentiable surrogate. Dynamic-security constraints will include phase-angle, frequency, and line-flow limits. Uncertainty in model parameters, fault duration, and stochastic forcing will be incorporated through scenario or chance-constrained formulations. Adjoint or sensitivity-based gradients will be used to efficiently evaluate the dependence of the objective and constraints on the decision variables [10]. A second stage will investigate learned parametric decision mappings for repeated optimization under changing operating conditions, following the learning-to-optimize framework [11], [3].

TABLE I: Two-year research plan and milestones.

	Year 1	Year 2
Learning to Model	Differentiable N1Plus implementation; EMT-data interface; baseline parameter identification	Topology-aware hybrid surrogate; validation across faults, durations, and operating points
Learning to Optimize	Formulation of differentiable dynamic-security objectives and constraints	Parametric optimization and fast decision mappings under uncertainty
Learning to Control	Differentiable predictive-control formulation	Contingency-aware model-predictive control and closed-loop evaluation
Deliverables	Open-source differentiable model; EMT benchmark; validation study	Optimization/control software; benchmark suite; final open-source release

Objective 3: Learning to Control. The third objective will use the same surrogate as a prediction model for closed-loop control. Unlike Objective 2, the control problem will determine a sequence of feedback actions during and after a disturbance. Initial studies will consider generator redispatch and controllable load adjustment over a finite prediction horizon. The control policy will be optimized through the differentiable surrogate while enforcing predicted state, flow, and actuator constraints. Uncertainty in the predicted dynamics will be incorporated through scenario-based or stochastic predictive control. The approach will build on differentiable predictive control methods that explicitly optimize control policies through differentiable dynamical models [4].

The two-year program will use measurable milestones. Months 1–6 will deliver the differentiable N1Plus implementation, EMT-data interface, and baseline parameter-identification study. Months 7–12 will deliver the graph-based hybrid surrogate and validation across fault locations, durations, and operating points. Months 13–18 will implement the differentiable dynamic-security optimization problems and quantify speedups relative to repeated EMT evaluation. Months 19–24 will implement the predictive-control formulation, evaluate it across uncertain contingencies, and release the complete open-source software and benchmark suite. Performance will be evaluated using predictive error, computational speedup, constraint violations, and generalization outside the calibration set.

V. Significance and Broader Impact

The proposed work addresses a central SciML problem: how to combine physical structure, data-driven learning, and differentiable computation when high-fidelity simulation is expensive. Transmission grids provide a demanding test case because their dynamics are governed by known network physics while remaining high-dimensional, uncertain, and subject to hard operational constraints.

The work has a direct methodological connection to Dr. Drgoňa's research agenda. NeuroMANCER provides a common differentiable framework for physics-informed system identification, constrained optimization, and model-based predictive control [3]. The proposed research will use the power-grid surrogate as a new testbed for this workflow: the same differentiable hybrid model will first be identified from EMT data, then embedded in dynamic-security optimization, and finally used as the prediction model for constrained control. The goal is therefore not simply to apply existing SciML tools to a new application, but to determine how a topology-aware, stochastic hybrid model can preserve domain structure while supporting these three differentiable tasks.

The connection to differentiable causal modeling is also methodological. The proposed surrogate separates known physical mechanisms from learned interactions and stochastic effects, creating an explicit computational structure in which individual components can be constrained, learned, and validated independently. This is consistent with differentiable causal block diagrams, where structured relationships between model components are retained while parameters and functional relationships remain differentiable [12]. For power grids, this structure provides a natural way to distinguish network physics, fault mechanisms, learned discrepancies, and uncertainty rather than treating the entire system as an unconstrained black box.

The broader outcome is a reusable methodology for differentiable simulation of constrained networked dynamical systems. The resulting software and benchmark problems will provide an open computational interface between high-fidelity simulation, system identification, optimization, and control. Although the primary application is transmission-grid dynamics, the same architecture is relevant to other systems in which physical models are available but expensive or incomplete.

VI. Anticipated Outcomes and Future Directions

The primary outcome will be a validated differentiable hybrid surrogate for transmission-grid dynamics, together with an open-source implementation and EMT benchmark. The first stage will establish whether learning only the unresolved component of the dynamics improves upon the existing stochastic surrogate while preserving its physical structure. Success will be evaluated using phase-angle RSE, trajectory accuracy, computational speedup, constraint violations, and generalization to fault locations and operating conditions outside the calibration set.

The resulting surrogate will then serve as a common differentiable model for two downstream tasks: dynamic-security optimization and model-based predictive control. The objective is to demonstrate that one physics-based learned representation can support prediction, decision-making, and control without repeatedly invoking high-fidelity EMT simulations.

Beyond the two-year project, the framework can be extended to larger networks, nonlinear and inverter-dominated dynamics, and richer uncertainty models. Differentiable causal representations provide a natural direction for composing these extensions while retaining explicit relationships between physical subsystems, learned components, and uncertainty [12]. The longer-term goal is therefore a reusable class of differentiable hybrid models for reliable simulation, optimization, and control of complex dynamical systems.

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